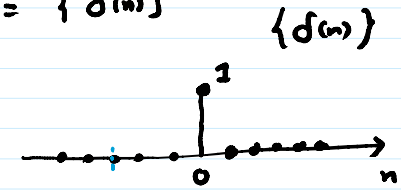
3. Moving Average System.

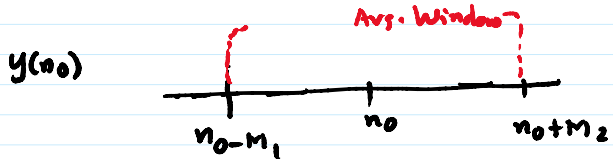
$$y(n) = \frac{\sum_{k=-M_1}^{M_2} x(n+k)}{M_1 + M_2 + 1}$$

Find impulse response $\{h(n)\}$

Fix $\{x(n)\} = \{\delta(n)\}$

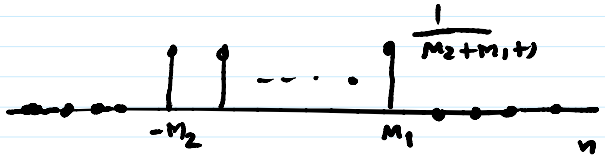


Find output at time index n_0



$$y(n_0) = \begin{cases} 0 \; ; & \text{if } n_0 + M_2 < 0 \\ 0 \; ; & \text{if } n_0 - M_1 > 0 \\ \frac{1}{M_1 + M_2 + 1} \; ; & \text{if } \left(\begin{array}{l} n_0 - M_1 \le 0 \\ \& \\ n_0 + M_2 \ge 0 \end{array}\right) \end{cases}$$

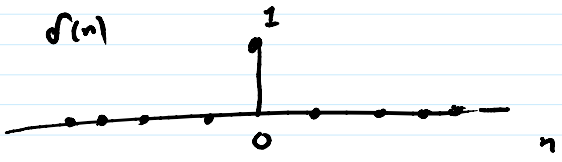
Imp Resp $\{h(n)\}$ =



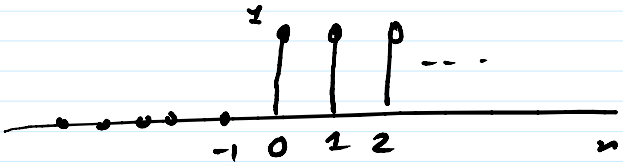
4. Accumulator.

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

Input $\delta(n)$



Output $h(n) = u(n)$



Finite Impulse Response (FIR) Systems

- Support of impulse response is a finite interval (length is finite)
- Example: moving avg, Time shift

Infinite Impulse response (IIR) Systems

- Size of support set of impulse response is infinite.
- Example: Accumulator

Example of Convolution:

$$\{x(n)\} = \{ \underset{\uparrow}{a}, b, c, d \} \quad n=0$$

$$\{h(n)\} = \{ \underset{\uparrow}{\alpha}, \beta, \gamma \}$$

$$\{h(n)\} = \{\underset{\downarrow}{\alpha}, \beta, \gamma\}$$

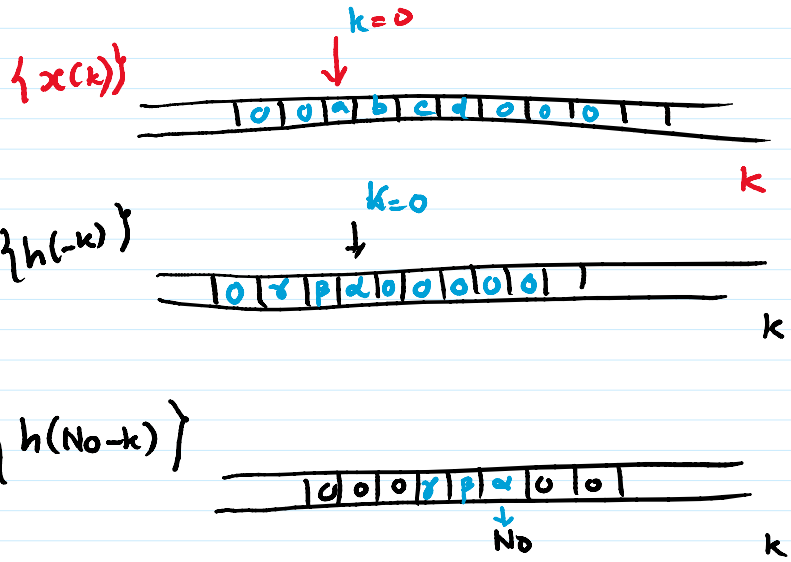
To find output (at a given time index N_0)

$$y(N_0) = \sum_{k=-\infty}^{\infty} x(k)\, h(N_0 - k)$$

- keep input $\{x(\cdot)\}$ as such

Flip & Slide Approach

- Time reverse & Time shift $\{h(\cdot)\}$



If $N_0 < 0$, $y(N_0) = 0$

$N_0 \geq 6$, $y(N_0) = 0$

$0 \leq N_0 \leq 5$; $y(N_0)$ will be non-zero

$\boxed{N_0 = 2}$ $\quad y(N_0) = \alpha \cdot c + \beta \cdot b + \gamma \cdot a$

Repeat for other values of N_0.

Properties of Convolution:

1. Delay Property.

Suppose $\{x(n)\} * \{h(n)\} = \{y(n)\}$

$\Downarrow$ Time shift by N_0 units

$$\{x(n-N_0)\} * \{h(n)\} = \{y(n-N_0)\}$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Time shifted $\{x_1(n)\} = \{x(n-N_0)\}$

Output $y_1(n) = \sum_{k=-\infty}^{\infty} \underbrace{x_1(k)}_{x(k-N_0)}\, h(n-k)$

$\downarrow$ $l = k - N_0$

$$y_1(n) = \sum_{l=-\infty}^{\infty} x(l)\, h(n-N_0-l)$$

$$y(n-N_0) = \sum x(k)\, h(n-N_0-k)$$

$$y(n-N_0) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-N_0-k)$$

$$\{y_1(n)\} = \{y(n-N_0)\}$$

2. Commutative

Suppose $\{x(n)\} * \{h(n)\} = \{y(n)\}$

Then $\{h(n)\} * \{x(n)\} = \{y(n)\}$

Proof: Change of Variables. method.

2a If $\{x(n)\} * \{h(n)\} = \{y(n)\}$

then $\{x(n)\} * \{h(n-N_0)\} = \{y(n-N_0)\}$

2c $\{x(n-N_1)\} * \{h(n-N_2)\} = \{y(n-(N_1+N_2))\}$

3 Associativity Property

Applicable only under some conditions on signals

on signals

Absolute Summability Condition:

DT Signal $\{x(n)\}$ is called absolutely summable

if $\sum_{n=-\infty}^{\infty} |x(n)| < \infty$

- If $\{x(n)\}$ has finite support
  then it will be absolutely summable always

- Infinite support

Absolutely summable.

- $x(n) = e^{-|n|}$
- $x(n) = 2^{-n} u(n)$

- Non absolutely summ

$x(n) = u(n)$

$x(n) = \cos(\omega_0 n)$

$x(n) = e^{j\omega_0 n}$